

```
GNU nano 7.2 p3t1.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;

    printf("==== Process Before fork() =====\n");
    printf("PID : %d\n", getpid());
    printf("PPID : %d\n", getppid());

    pid = fork();

    if (pid < 0)
    {
        printf("Fork failed!\n");
        exit(1);
    }

    else if (pid == 0)
    {
        // Child Process
        printf("\n==== Child Process =====\n");
        printf("Child PID : %d\n", getpid());
        printf("Parent PID : %d\n", getppid());

        printf("Child is running...\n");
        sleep(5);

        printf("Child process terminating...\n");
        exit(0);
    }

    else
    {
        // Parent Process
        printf("\n==== Parent Process =====\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);

        printf("Parent is waiting for child...\n");

        wait(NULL);

        printf("Child process completed.\n");
        printf("Parent process terminating...\n");
    }

    return 0;
}
```

```
gopinath@GOPINATH:~$ nano p3t1.c
gopinath@GOPINATH:~$ gcc p3t1.c -o p3t1
gopinath@GOPINATH:~$ ./p3t1
==== Process Before fork() =====
PID : 1523
PPID : 320

==== Parent Process =====
Parent PID : 1523
Child PID : 1524
Parent is waiting for child...

==== Child Process =====
Child PID : 1524
Parent PID : 1523
Child is running...
Child process terminating...
Child process completed.
Parent process terminating...
gopinath@GOPINATH:~$
```

TASK-2

```
GNU nano 7.2 p3t2.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;

    pid = fork();

    if (pid < 0)
    {
        printf("Fork failed!\n");
        exit(1);
    }

    else if (pid == 0)
    {
        // Child Process
        printf("Child Process Started\n");
        printf("Child PID : %d\n", getpid());
        printf("Parent PID : %d\n", getppid());

        printf("Child is running...\n");

        // Child enters sleeping/waiting state
        printf("Child is going to sleep...\n");
        sleep(20);

        printf("Child woke up and is running again...\n");

        sleep(5);

        printf("Child terminating...\n");
        exit(0);
    }

    else
    {
        // Parent Process
        printf("Parent Process Started\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);

        printf("Parent is waiting for child...\n");

        // Parent waits for child
        wait(NULL);

        printf("Child terminated.\n");
        printf("Parent terminating...\n");
    }

    return 0;
}
```

```
gopinath@GOPINATH:~$ nano p3t2.c
gopinath@GOPINATH:~$ gcc p3t2.c -o p3t2
gopinath@GOPINATH:~$ ./p3t2
Parent Process Started
Parent PID : 1566
Child PID : 1567
Parent is waiting for child...
Child Process Started
Child PID : 1567
Parent PID : 1566
Child is running...
Child is going to sleep...
Child woke up and is running again...
Child terminating...
Child terminated.
Parent terminating...
gopinath@GOPINATH:~$ ps -ef | grep task
gopinath      1571      320  0 13:27 pts/0    00:00:00 grep --color=auto task2
gopinath@GOPINATH:~$ ps -o pid,ppid,state,stat,cmd -p <PID>
-bash: syntax error near unexpected token `newline'
gopinath@GOPINATH:~$ ps -o pid,ppid,state,stat,cmd -p 2451
  PID   PPID S  STAT CMD
gopinath@GOPINATH:~$ cat /proc/2451/status
cat: /proc/2451/status: No such file or directory
gopinath@GOPINATH:~$ |
```